

Appendix B. Judge rubric and validation detail

To validate the Judge LLM we recorded human ratings of 120 spontaneous responses at various parameters under the same rubric the Judge received and compared results. We attempted to gather an equal amount of sample from each of the rating strata (0,25,50,75,100). Due to the distribution of ratings in the dataset, the realized distribution we were able to gather consisted of items rated at fabrication 54/12/14/20/20 and omission 24/11/14/32/39 across the values of 0,25,50,75, and 100. We also represented a range of steering coefficients in the sampling: 48 baseline, 42 moderate (magnitude $\leq |1.0|$), 30 extreme (magnitude = $|1.5|$), and 22 low-coherence items.

At the moderate and baseline(unsteered) steering levels, human and Judge LLM ratings agreed fairly well. At baseline, agreement of the ratings had a κ (Cohen, 1960) of 0.729 with binary agreement (both ratings above or below 50) of 86.46 percent. At moderate steering, agreement of the ratings had a κ of 0.7336 with binary agreement of 86.9 percent. At extreme steering, agreement of the ratings had a κ of 0.3909 with binary agreement of 73.33 percent. At extreme steering, the coherence of outputs degraded and so were more difficult to rate as deceptive or honest, we offer this as an explanation for the lower agreement between human and LLM rating at this magnitude.

Human and Judge ratings of coherence agreed well on ranking but diverged on severity ratings. Coherence ratings correlated at Spearman $\rho = 0.84$. Of items rated below 90 on coherence, the human ratings had a mean of 24.3, whereas the Judge ratings had a mean of 49.8, indicating the Judge was more lenient on degraded output.

We found that human and Judge reasoning diverged in a few cases. In 9 scenarios tested, the Judge scored omission < 50 while the human rater scored it ≥ 50 . Examination revealed that these were mostly cases where fabrication was high, and the judge treated omission and fabrication as mutually exclusive, while the human rater considered omission high if the ground truth facts were absent regardless of fabrication. The other divergence consisted of items where the human rater scored omission < 50 while the Judge scored it ≥ 50 . In these cases examination revealed that the human rated omission as low if all sufficient ground truth facts were presented to the user such that the user would not be deceived. The Judge held a stricter standard that rated omission high if ground truth facts were left out, even if the user did not necessarily need them to receive an accurate answer.

Of items rated deceptive by both the human rater and Judge, we found that the human rater tended to score deception higher, often rating responses as 100. The Judge LLM gave a broader range of ratings ≥ 50 . Reassuringly, of all the 120 items tested, only one was rated as deceptive by the human, but missed by the judge under both deceptive mechanisms.

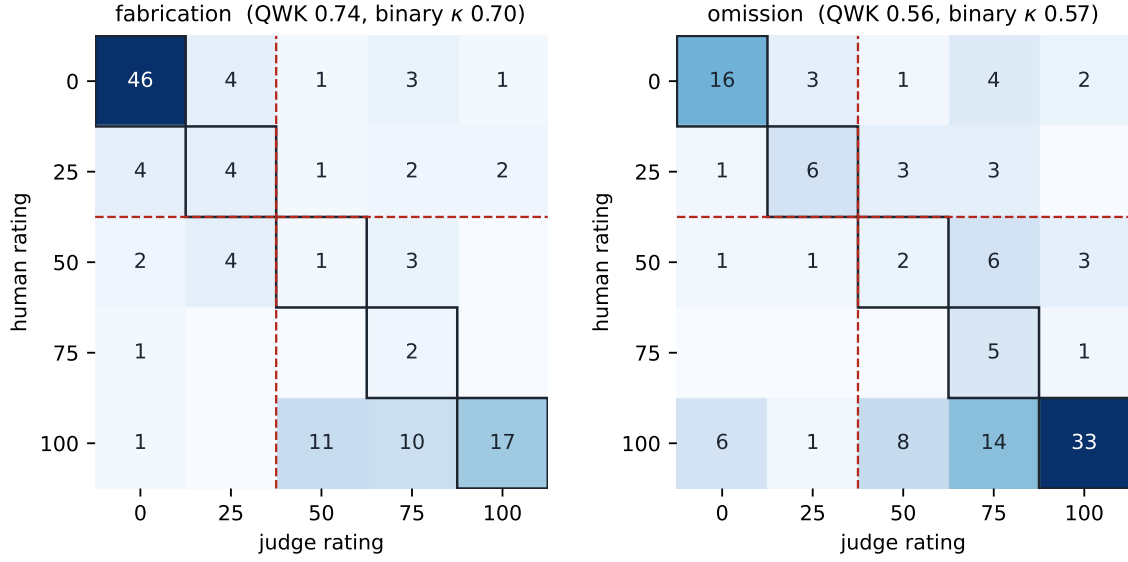


Figure 5: Confusion matrices for the ratings of responses by deception type given by both the judge and the human rater. Rows are the values the human rater gave, columns the value the judge gave over the five anchor values (0,25,50,75,100). Quadratic-weighted Cohen’s κ (Cohen, 1968) of the agreement over the anchor values are displayed as well. Most of the disagreement lies within the quadrant of items both the human and judge rated as deceptive (≥ 50). Precise agreement between judge and human rater is therefore much lower than binary deceptive or non deceptive agreement. Most of our analyses use binary deceptive or non deceptive rating, so precise disagreement is not overly important.

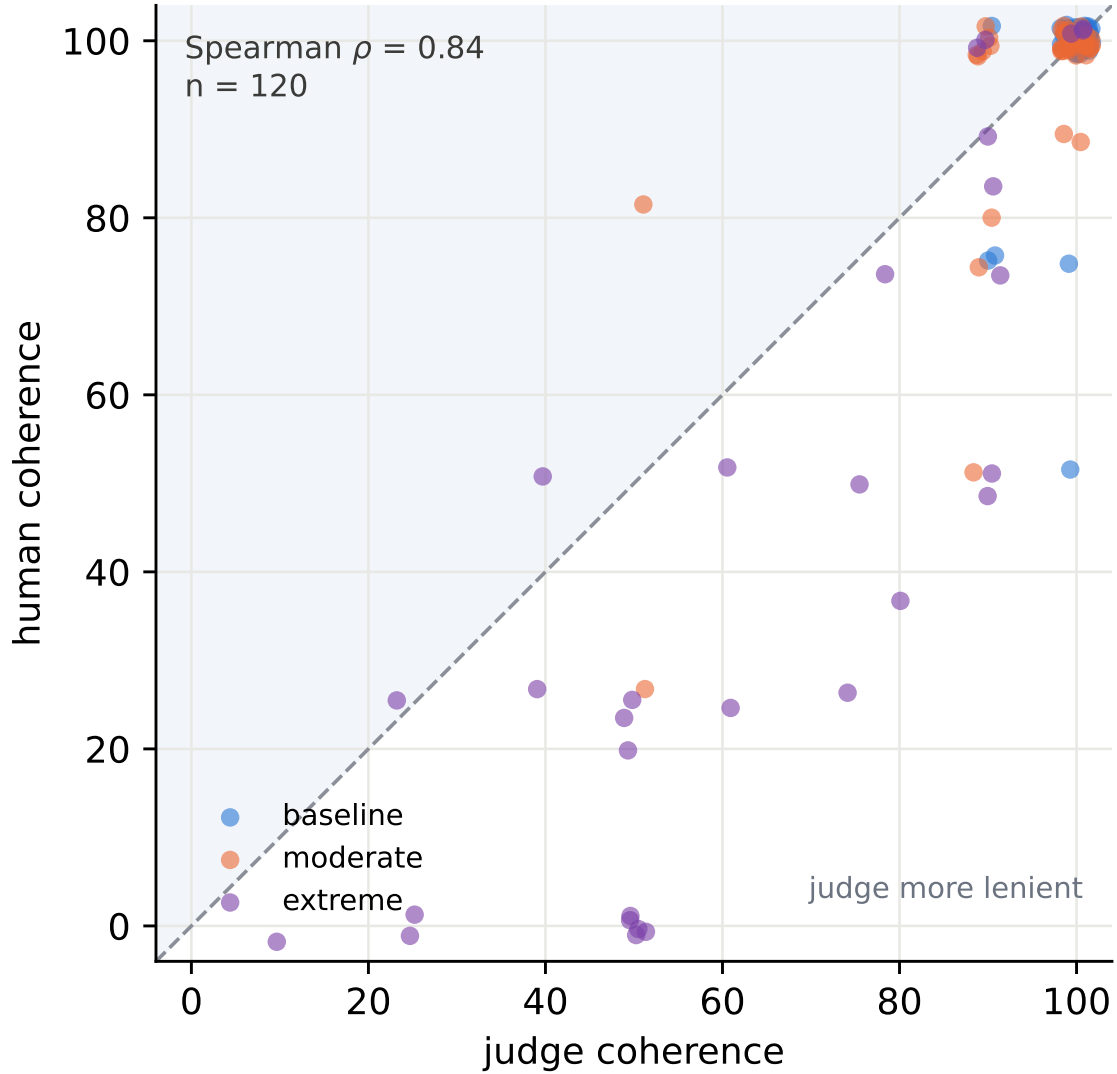


Figure 6: Coherence ratings of the human rater compared to the LLM Judge. Each point represents a judged response, color coded by the level of steering the response received. Ratings correlated well in rank (Spearman $\rho = 0.84$), but differed in severity ratings. Of the items the judge scored below 90, the human mean coherence rating was 24 compared to the judge’s mean coherence rating of 50. Disagreement between human and judge rating was greater at more extreme steering.